

IoT-based Water Quality Monitoring Station and Forecasting System with Machine Learning

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Abstract—This paper presents an IoT-based water quality monitoring and forecasting system designed for real-time and continuous assessment of water resources. The system integrates Siemens SIMATIC IOT2050 as an Industrial IoT Gateway, which collects data from sensors measuring conductivity, pH, dissolved oxygen, and temperature using RS485 Modbus RTU communication. Data processing occurs at the edge using Node-RED and is transmitted to AWS Cloud via MQTT for storage and visualization on a dashboard. Predictive analysis employs machine learning models, including XGBoost with Optuna parameter tuning and Long Short-Term Memory (LSTM) networks, for water quality forecasting. Results indicate superior performance of LSTM for most parameters, while XGBoost excels in pH prediction. This system demonstrates scalability, reliability, and potential for enhanced water quality management in diverse environments.

Keywords—Internet of Things, Water Quality Monitoring, MQTT, LSTM

I. INTRODUCTION

Water quality monitoring plays a pivotal role in managing water resources and ensuring environmental sustainability. As water is a fundamental resource for human life and ecosystems, maintaining its quality to meet established standards is essential for mitigating health risks, preserving ecological balance, and fostering sustainable development. Traditional water quality monitoring methods, often reliant on manual assessments, are not only resource-intensive but also constrained by limitations in speed and accuracy of data reporting. Additionally, the accessibility of timely and precise water quality information remains a significant challenge. The integration of digital technologies has emerged as a transformative approach to address these challenges in water quality monitoring. Among these technologies, the Internet of Things (IoT) [1] has demonstrated substantial potential in providing efficient and scalable solutions for diverse applications, including smart farming [2], smart cities [3], smart healthcare [4], and smart aquaculture [5]. IoT technology facilitates the deployment of sensors for real-time data collection, minimizes delays, and enhances accuracy in monitoring processes. This is achieved through wireless communication to cloud servers, enabling data visualization via real-time dashboards that present information in an intuitive format. Several studies have explored IoT-based approaches for water quality monitoring. For instance, T. Anuwongpinit et al. [6] developed a portable IoT device using microcontroller technology for monitoring saline water quality in the Bang Kachao area of Thailand. However, this system was limited by its reliance on Wi-Fi connectivity, resulting in intermittent data availability and challenges in

continuous operation. Similarly, U. Seeboonruang et al. [7] developed an automated alert system for salinity monitoring using a dedicated water quality station. While effective for real-time salinity monitoring, the system faced constraints in its measurement capabilities and power supply options, leading to data loss and limited applicability for broader water quality parameters. Additional studies [8–11] proposed water quality monitoring systems based on microcontroller boards, but these too were hampered by limitations in continuous operation and compatibility with sustainable power sources such as solar energy. To address these limitations, this research proposes the design and development of an advanced IoT-based water quality monitoring station. The system is capable of measuring multiple water quality parameters, including electrical conductivity, pH levels, dissolved oxygen, and water temperature. The collected data is visualized through a web-based application, offering real-time graphical representations for enhanced accessibility and decision-making. Furthermore, the system incorporates predictive analysis capabilities by employing machine learning techniques, including multi-level linear regression and the Extreme Gradient Boosting (XGBoost) model. To enhance forecasting accuracy, parameter tuning is performed using Optuna. In addition, the study explores the application of a Long Short-Term Memory (LSTM) model to evaluate the effectiveness of deep learning techniques in forecasting water quality parameters. A comparative analysis of these models is conducted to determine their practical suitability for real-world applications..

II. SYTEM DESIGN AND ARCHITECTURE

A. System Overview and Architecture

The overall design of the development of a water quality monitoring station using the Internet of Things (IoT) consists of two parts: the design of the Amazon Web Services (AWS) Cloud and the design of the water quality monitoring station. The overall overview of this research is presented in Fig. 1.

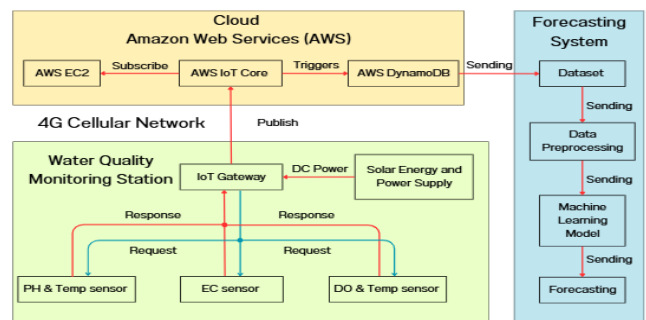


Fig. 1. Overview of the proposed system design

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In the proposed system, data from the water quality monitoring station is seamlessly integrated with AWS Cloud services using AWS IoT Core. The connection is established via the Message Queuing Telemetry Transport (MQTT) protocol, which enables efficient data exchange. A broker within AWS IoT Core serves as an intermediary, facilitating the receipt and transmission of data between the water quality monitoring station and cloud services. The broker triggers functions to store the sensor data into an AWS DynamoDB database for persistent storage. For data visualization, AWS EC2 acts as the server hosting a dashboard that displays real-time and historical data obtained from the sensors. This dashboard subscribes to data streams from the AWS IoT Core broker using the MQTT protocol, ensuring the presentation of accurate and up-to-date information. At the water quality monitoring station, various sensors collect key environmental parameters and transmit the data to a Siemens SIMATIC IOT2050 Industrial Internet of Things (IIoT) Gateway via the RS485 Modbus RTU serial communication protocol. The IOT2050 gateway performs Edge Computing, processing the sensor data locally before publishing it to AWS Cloud via the MQTT protocol over a 4G cellular network. The water quality monitoring station is powered by a dual energy system, utilizing both solar energy and a conventional power supply to ensure continuous operation. This robust energy strategy enhances reliability and supports the station's deployment in remote or off-grid locations.

B. Hardware Design

The hardware system design of the water quality monitoring station consists of three parts: the power source, IIoT Gateway, and sensors. An overview of the hardware system of the station is shown in Fig. 2 and Fig. 3.

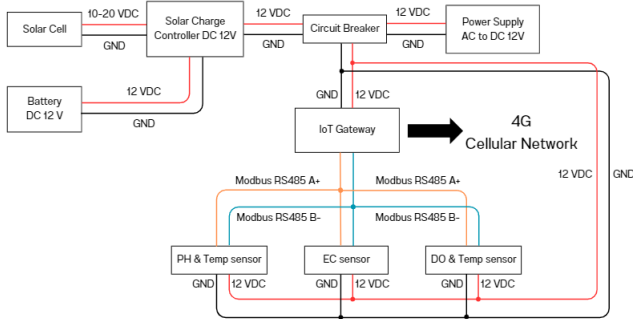


Fig. 2. Overview of the station's hardware system



Fig. 3. Example of designed station

The diagram illustrates the data transmission between the IIoT Gateway and the sensors via the RS485 Modbus RTU standard, which utilizes only two data lines: RS485 A+ and RS485 B-. These lines connect each sensor to the IIoT Gateway. Data from the IIoT Gateway is sent to the AWS Cloud through the internet using a 4G cellular network

integrated within the IIoT Gateway, employing the MQTT Protocol to publish the data to the AWS IoT Core server broker. Additionally, the power supply for the system is designed to maintain a constant voltage of 12 voltage, derived from two sources: power from a Power Supply, which converts alternating current (AC) to direct current (DC), and energy from Solar Energy. The system includes a Solar Charge Controller to regulate the power supply, receiving energy from solar cells and maintaining a stable voltage to charge a battery for backup power when solar energy is not available. This setup also powers the control cabinet system, with a circuit breaker acting as a switch to control the electricity from both power sources.

C. Software Design

The software design within the IIoT Gateway will utilize development tools for device connectivity, specifically Node-RED [12], which operates on Node.js within the Debian (Linux) operating system. The program will initiate by reading data from each sensor through the RS485 Modbus RTU standard, with each sensor's configuration with 9600 baud rate and specified function code, slave id and data address. The configuration of parameters for each type of sensor, ensures that data does not overlap within the RS485 Modbus RTU standard. This includes details regarding data transmission, such as the transmission rate, address, and functions for accessing the data, as well as the voltage requirements for each sensor.

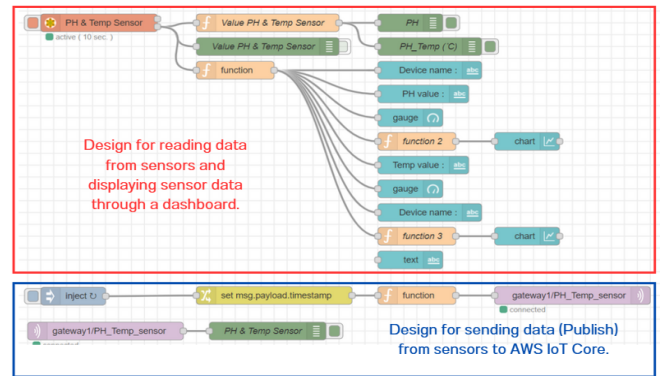


Fig. 4. System overview with Node-RED in IIoT Gateway

The overview of the software design for the system and management using Node-RED within the IIoT Gateway is illustrated in Figure 4. The red-bordered section represents the reading of data from each sensor, configured through the Modbus-Read Node and connected to a Function Node to write functions and process the data for display on the IIoT Gateway dashboard. The blue-bordered section indicates the transmission (publish) of data to the AWS Cloud IoT Core server broker. This is achieved by connecting to a Set Node and a Function Node to write functions and configure the data transmission to the MQTT-Out Node, which sends data to the AWS Cloud IoT Core server broker via the MQTT Protocol using the internet over a 4G cellular network.

The overview of the software design for the system and management using Node-RED within AWS Cloud EC2 is illustrated in Figure 5. The program begins by reading (subscribing to) data from each sensor from the AWS Cloud IoT Core server broker via the MQTT Protocol. This is configured through the MQTT-In Node and connected to a Function Node to write functions and process the data for display on the dashboard in AWS Cloud EC2.

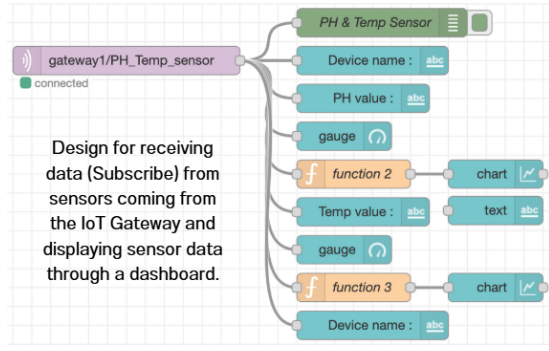


Fig. 5. System overview with Node-RED within AWS EC2

D. Sensor Calibration

The calibration of measurement instruments is essential for enhancing data accuracy and involves three sensors: the electrical conductivity (EC) sensor, the pH sensor, and the temperature sensor. For the EC sensor, the calibration begins with cleaning and thoroughly drying the probe, followed by preparing a standard conductivity solution of 1413 $\mu\text{S}/\text{cm}$. The calibration process is initiated by pressing and holding the calibration button until the yellow light appears, releasing it, and pressing the button twice until the red-light blinks. Once the probe is immersed in the solution and the calibration button is pressed once, the process completes when the light turns off. For the pH and temperature sensors, calibration involves cleaning the sensors and preparing buffer solutions of pH 4.00 and pH 9.18. The pH 4.00 calibration starts by immersing the sensors in the solution, holding the calibration button until the yellow light appears, pressing the button twice until the red-light blinks, and waiting 30 seconds until the light turns off. After rinsing and drying the sensors, they are immersed in the pH 9.18 solution, and the calibration button is pressed once to activate a green blinking light for 30 seconds, followed by a 20-second wait as the light changes to blinking yellow, signaling completion.

III. FORECASTING SYSTEM DESIGN

The design of the water quality forecasting model for the Chao Phraya River time series compares the LSTM model and the XGBoost model. This aims to enhance the ability to monitor water quality continuously and comprehensively across various locations. The parameters used include date and time, water temperature, water conductivity, dissolved oxygen levels, and acidity-alkalinity (pH). The LSTM model, which is based on the principles of Recurrent Neural Networks (RNN) and falls under the category of deep learning, is designed to remember patterns in data over extended periods, making it effective for forecasting sequential data. Meanwhile, the XGBoost model employs decision tree principles, learning from sequential data across multiple trees, where each decision tree learns from the errors of the previous tree, thereby increasing forecasting accuracy. This study compares the models to identify which one yields the best forecasting results, making it suitable for practical applications.

A. Dataset

The study used water quality data from the Chao Phraya River at the Makham station over the past 10 years [13], provided by the Metropolitan Waterworks Authority. The parameters included date and time, water temperature,

electrical conductivity, total dissolved solids, salinity, dissolved oxygen content, depth, pH level, chlorophyll concentration, and water turbidity, respectively. The sample of water quality data from the Chao Phraya River at the Makham station presented in this study is shown in Fig. 6.

date_time	Temper...	Conduct...	TDS (m...	Salinity ...	DO (mg/L)	Sensor ...	pH	Chlorop...	Turbid (...)
2014-07-05T23:20:00	31.59	330	210	0.16	4.54	1.25	7.15	13.3	17.4
2014-07-05T23:30:00	31.6	330	220	0.16	4.44	1.26	7.15	12.1	18.9
2014-07-05T23:40:00	31.6	333	220	0.16	4.44	1.26	7.14	12.1	16.7
2014-07-05T23:50:00	31.6	334	220	0.16	4.27	1.27	7.11	12.6	15.6
2014-07-06T00:00:00	31.62	338	220	0.16	4.25	1.28	7.11	14	16.9
2014-07-06T00:10:00	31.61	340	220	0.16	4.15	1.28	7.1	13.9	17.9
2014-07-06T00:20:00	31.6	341	220	0.16	3.99	1.29	7.07	13	15.9
2014-07-06T00:30:00	31.59	345	220	0.16	3.92	1.29	7.06	14.4	18.5
2014-07-06T00:40:00	31.57	346	230	0.16	3.8	1.3	7.05	15.1	18
2014-07-06T00:50:00	31.56	348	230	0.16	3.62	1.3	7.03	14	17.7
2014-07-06T01:00:00	31.54	349	230	0.16	3.49	1.31	7.01	13.9	16
2014-07-06T01:10:00	31.53	351	230	0.17	3.35	1.31	7	13.1	16.7
2014-07-06T01:20:00	31.52	354	230	0.17	3.24	1.31	6.99	13.4	18.1
2014-07-06T01:30:00	31.51	357	230	0.17	3.15	1.31	6.98	13.6	15.4
2014-07-06T01:40:00	31.5	358	230	0.17	3.04	1.32	6.97	13.6	15.1
2014-07-06T01:50:00	31.5	362	240	0.17	2.97	1.32	6.96	13.2	16.4
2014-09-30T03:30:00	0	0	0	0	0	0	0	0	0
2014-07-06T02:00:00	31.49	365	240	0.17	2.91	1.32	6.96	13.3	14.9
2014-07-06T02:10:00	31.49	366	240	0.17	2.89	1.32	6.95	12.9	14.3
2014-07-06T02:20:00	31.49	371	240	0.18	2.86	1.32	6.95	13	14.6
2014-07-06T02:30:00	31.49	373	240	0.18	2.79	1.32	6.94	13.7	15.2

Fig. 6. Water quality data of Chao Phraya River at Wat Ma-kham station

The data cleaning process began by removing parameters that were not studied, which included total dissolved solids, salinity, depth, chlorophyll concentration, and water turbidity. This left only the following parameters: date and time, water temperature, electrical conductivity, dissolved oxygen content, and pH level. It also involved checking for impossible data values, such as 0, empty values, negative values, and values that exceed realistic limits. For missing data, interpolation was used by applying the average of the preceding and succeeding data points, which resulted in a dataset ready for LSTM and XGBoost model analysis. The data after formatting is presented in Fig. 7.

date_time	Temperature\n(C)	Conductivity\n(uS/cm)	DO\n(mg/L)	pH
2014-04-28	29.038523	501.199898	7.757047	9.276510
2014-04-29	29.433154	501.199898	7.346309	9.045638
2014-04-30	30.001423	501.199898	6.754846	8.713181
2014-05-01	30.569691	501.199898	6.163383	8.380725
2014-05-02	31.137960	501.199898	5.571919	8.048268
...	...	...	...	...
2024-07-21	31.365139	351.027778	2.146181	3.080000
2024-07-22	31.287986	345.090278	2.114375	3.080000
2024-07-23	31.137361	338.798611	2.126111	3.080000
2024-07-24	30.967847	322.312500	2.622986	3.080000
2024-07-25	30.821606	288.642336	3.139635	3.080000

Fig. 7. Water quality data after data cleaning process

In addition, each model utilizes different data formatting techniques. For LSTM, the dataset is created to establish a relationship between x and y . In contrast, XGBoost incorporates seasonal relationships or holidays and includes time-based interactions to enhance the model's ability to make accurate predictions. The data is then sorted chronologically, with the values normalized to a range of 0–1 (Min-Max Normalization).

B. Long Short-Term Memory

Long Short-Term Memory (LSTM) [14] networks address the vanishing gradient problem in standard recurrent neural networks (RNNs) by introducing memory cells and gating mechanisms. These mechanisms enable LSTMs to

capture long-term dependencies in sequential data. The Mathematical Framework is as follows:

LSTM Cell Structure: Each LSTM cell consists of three gates and a cell state:

- Forget Gate: Discards irrelevant information from the previous cell state.
- Input Gate: Adds new relevant information to the cell state.
- Output Gate: Determines the output based on the updated cell state.

For the gate equations, let x_t be the input at time t , h_{t-1} the hidden state from the previous time step, and c_{t-1} the previous cell state.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (1)$$

Where σ is the sigmoid activation function, W_f is the weight matrix, and b_f is the bias vector.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (2)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (3)$$

Where i_t controls how much new information to add, and $\tilde{c}_t$ is the candidate cell state.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (4)$$

$$h_t = o_t \odot \tanh(c_t) \quad (5)$$

Recursive Flow: The cell state c_t acts as memory, while h_t represents the output at each time step. The recursive flow is essential for processing sequential data. LSTMs ensure stable gradient flow through the memory cell, enabling effective learning of long-term dependencies.

C. Extreme Gradient Boosting Model Design

Extreme Gradient Boosting (XGBoost) [15] is a highly efficient and scalable implementation of the gradient boosting framework. It is designed to optimize both computation time and predictive performance. The method constructs an ensemble of decision trees by iteratively adding new models to correct errors from previous iteration. The Mathematical Framework is as follows: Boosting is an additive learning process where a weak learner is iteratively added to improve model predictions. Let the dataset be $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ where x_i represents the input features and y_i the target values. The model prediction at step t is (6).

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i), \quad f_k \in F \quad (6)$$

Where $f_k(x_i)$ is the k -th decision tree (weak learner) and F is the functional space of decision trees.

Objective Function: The objective function balances the loss function and model complexity as (7).

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^t \Omega(f_k) \quad (7)$$

Where $l(y_i, \hat{y}_i)$ is loss function (e.g., mean squared error for regression or log loss for classification) and $\Omega(f_k)$ is regularization term for controlling model complexity.

Regularization: Regularization penalizes the complexity of individual trees to prevent overfitting:

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (8)$$

Where T is number of leaves in the tree, w_j is weight of the j -th leaf, γ is minimum loss reduction required to add a leaf node and λ is L2 regularization coefficient.

Gradient Boosting Approximation: To optimize the objective function, XGBoost uses a second-order Taylor expansion of the loss function:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \quad (9)$$

Where $g_i = \frac{\partial l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}$ is first-order gradient and

$h_i = \frac{\partial^2 l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)2}}$ is second-order gradient.

Tree Splitting and Gain Calculation: At each tree node, the split is chosen to maximize the reduction in the loss function (gain). For a split that divides the data into I_L (left) and I_R (right) subsets, the gain is:

$$\text{Gain} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} + \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma \quad (10)$$

The split that yields the maximum gain is chosen.

Prediction Update with Shrinkage: After each tree is built, the model prediction is updated using a learning rate η to reduce overfitting:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(x_i) \quad (11)$$

Where $0 < \eta \leq 1$

IV. EXPERIMENTAL RESULTS

A. Water Quality Station Operation

Operation of the water quality monitoring station begins by reading data (subscribing) from the AWS IoT Core Server Broker via the MQTT protocol in the cloud. The process starts with the DO and temperature sensor, followed by the pH and temperature sensor, and then the EC sensor, in that order. The data consists of a timestamp, device name, and data, respectively, formatted as JavaScript Object Notation (JSON), which is a standard for data interchange. shown in Fig. 8-9.



Fig. 8. Operation of the station

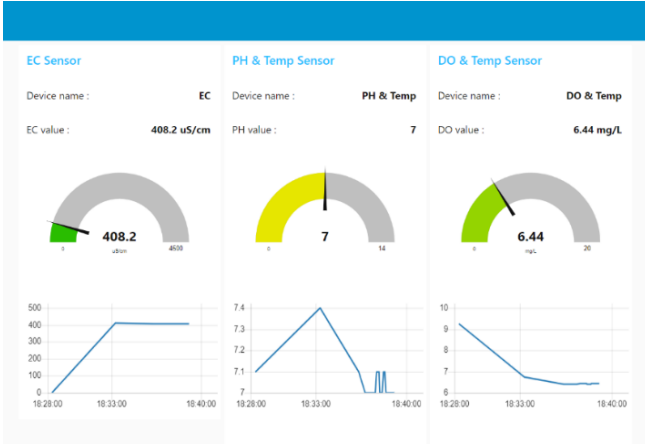


Fig. 9. Dashboard showing water quality of the station

Sensor data collected via the RS485 Modbus RTU standard is transmitted from the Industrial Internet of Things (IIoT) gateway to the AWS IoT Core server broker using the MQTT protocol over a 4G cellular internet connection. The transmitted data includes a timestamp, device identifier, and measurement values, formatted in JavaScript Object Notation (JSON), a widely used standard for data interchange. Upon reaching AWS IoT Core, the data is stored in an AWS DynamoDB database through trigger functions that automate the writing process, ensuring seamless integration of sensor data from the control cabinet into the database. This stored information is subsequently visualized on a dashboard hosted on AWS EC2, which operates within the Amazon Linux environment. The dashboard development utilizes Node-RED, running on Node.js, to enable device connectivity and data visualization. The system processes begin with subscribing to the sensor data streams from the AWS IoT Core server broker via the MQTT protocol, followed by executing programmed functions to process and display the data effectively on the cloud-hosted dashboard.

B. Water Quality Prediction and Testing Methods

The time series dataset on water quality of the Chao Phraya River at the Macam Station covers the past 10 years and is sourced from the Metropolitan Waterworks Authority. The parameters used for forecasting in this study's model include water temperature, water conductivity, dissolved oxygen levels, and acidity-alkalinity (pH).

The performance measurement parameters for the model include the Coefficient of Determination Score (R-Squared), which indicates the percentage of variability in the response variable that can be explained by the linear model, represented as the explained variance divided by the total variance. A higher R-Squared value, closer to 1, indicates better model performance. If it equals 1, it means the model predicts values with 100% accuracy. However, in practice, achieving such accuracy through model training is quite challenging, as shown in (12).

$$R^2 = 1 - \left(\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \right) \quad (12)$$

Mean Squared Error (MSE) is a method for measuring the difference between predicted values and actual values. It is calculated by determining the squared errors and then

finding their average. A lower MSE value, closer to 0, indicates better model performance, as shown in (13).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (13)$$

Mean Absolute Error (MAE) is a method for measuring the difference between predicted values and actual values. It is calculated by taking the absolute differences between the predicted values and the actual values for each instance, summing them up, and averaging the result. A lower MAE value, closer to 0, indicates better model performance, as shown in Equation (14).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (14)$$

Root Mean Squared Error (RMSE) is a method for measuring the difference between predicted values and actual values. It is calculated by taking the square root of the average of the squared errors. A lower RMSE value, closer to 0, indicates better model performance, as shown in Equation (15).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|^2} \quad (15)$$

The process of training and testing each model begins with using 70% of the entire dataset for each parameter: water temperature, water conductivity, dissolved oxygen levels, and acidity-alkalinity (pH). The remaining 30% of the dataset is then used for testing. The performance of each model is compared to determine which parameter provides better forecasting accuracy, with the performance metrics displayed in table 1.

TABLE I. THE PREDICTIVE PERFORMANCE OF THE FOUR PARAMETERS OF BOTH MODELS.

Efficiency	Temperature		Electrical Conductivity		Dissolved Oxygen		pH	
	LSTM	XGBoost	LSTM	XGBoost	LSTM	XGBoost	LSTM	XGBoost
R-Squared	0.8854	0.8442	0.8433	0.766	0.8946	0.7594	0.7485	0.5111
MAE	0.0263	0.0315	0.0119	0.0193	0.0431	0.0604	0.022	0.011
RMSE	0.0444	0.0518	0.0237	0.0289	0.0624	0.094	0.0403	0.0399
MSE	0.0019	0.0026	0.0005	0.0008	0.0038	0.0089	0.0016	0.0015

From table 6, it can be seen that for forecasting the parameters of water temperature, water conductivity, and dissolved oxygen levels, the LSTM model has R-Squared values that are closer to 1 compared to the XGBoost model. Additionally, the MAE, RMSE, and MSE values for the LSTM model are all closer to 0 than those of the XGBoost model. Therefore, the forecasting of water temperature, water conductivity, and dissolved oxygen levels using the LSTM model is better than that using the XGBoost model.

In contrast, for forecasting the acidity-alkalinity (pH) parameter, the LSTM model also has an R-Squared value closer to 1 than the XGBoost model, but the MAE, RMSE, and MSE values for the XGBoost model are closer to 0 than those of the LSTM model. Thus, the forecasting of the acidity-alkalinity parameter using the XGBoost model is better than that using the LSTM model.

V. CONCLUSION

This system can be applied to continuously and comprehensively monitor and check water quality in various areas, enhancing the accuracy of analysis and reducing delays in data collection for effective water quality management. The focus is on developing water quality monitoring stations that can operate continuously, saving time and providing user-friendly convenience through cellular network connectivity. This includes sending data to websites that facilitate efficient water quality monitoring and assessment. Additionally, the study explores the development of predictive systems for future water quality monitoring using multi-level linear regression models and deep learning with neural networks, employing time-series data from the Chao Phraya River over the past 10 years from the Metropolitan Waterworks Authority.

The results indicate that the LSTM model can predict time-series data for three parameters: water temperature, water conductivity, and dissolved oxygen levels, with performance metrics including R-Squared, MAE, RMSE, and MSE consistently showing that the LSTM model outperforms the XGBoost model. Conversely, the XGBoost model effectively predicts time-series data for the acidity-alkalinity parameter, with similar performance metrics indicating that it performs better than the LSTM model.

For future research development, the system will be enhanced by adding additional sensors to measure and monitor a variety of water quality parameters, such as sensors for total dissolved solids, inorganic and organic substances in water, turbidity sensors, and salinity sensors. This will enable each station to assess a wider range of water quality parameters that may affect water quality and serve as a model for applications in other areas. Moreover, the system will include features for alerts and data display on dashboards through mobile applications to ensure quick and easy access to data visualization, allowing for ongoing and seamless use. The data obtained from the water quality monitoring stations will be utilized for future water quality prediction.

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